Advanced Materials

Soft Matter 2023

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Bitte verzichten Sie in dieser Lehrveranstaltung
auf Bild- und Tonaufzeichnungen.

Weitere Informationen finden Sie im Studierendenportal

Das Veröffentlichen oder Teilen von Bild- und Tonaufzeichnungen dieser Lehrveranstaltung ist nicht gestattet.

Weitere Informationen finden Sie im Studierendenportal

Dies gilt auch für Lehrmaterial wie die Folien zur Vorlesung.

Inhaltsverzeichnis:

Anwendung von Photoinitiatoren

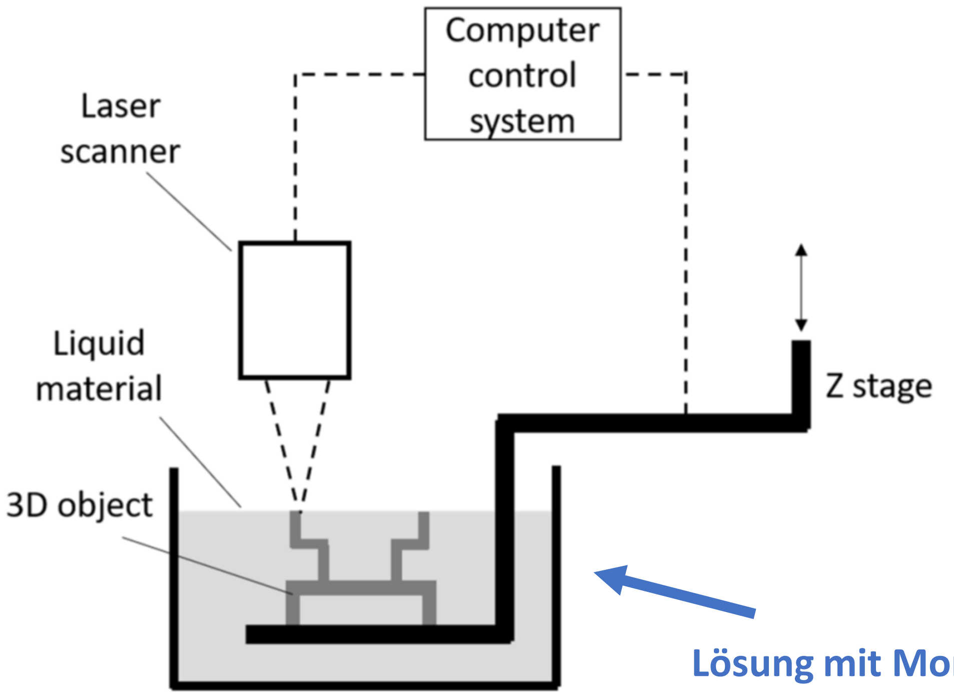
*3D-Druck

*Zahnfüllung

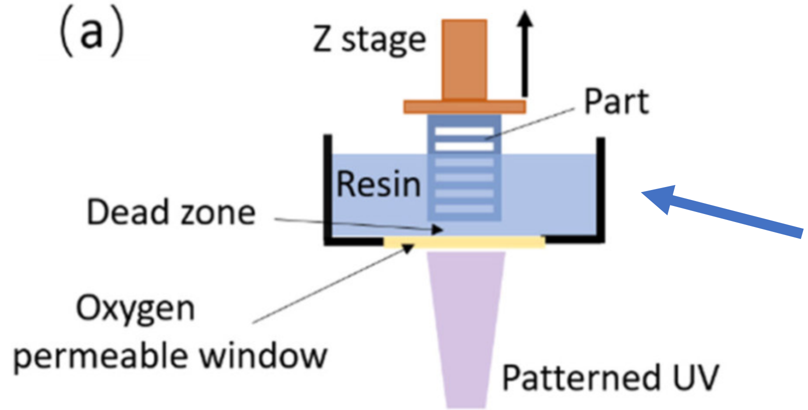
*Kontaktlinsen

Stereolitography (3D Sruck)

Bottom-Up



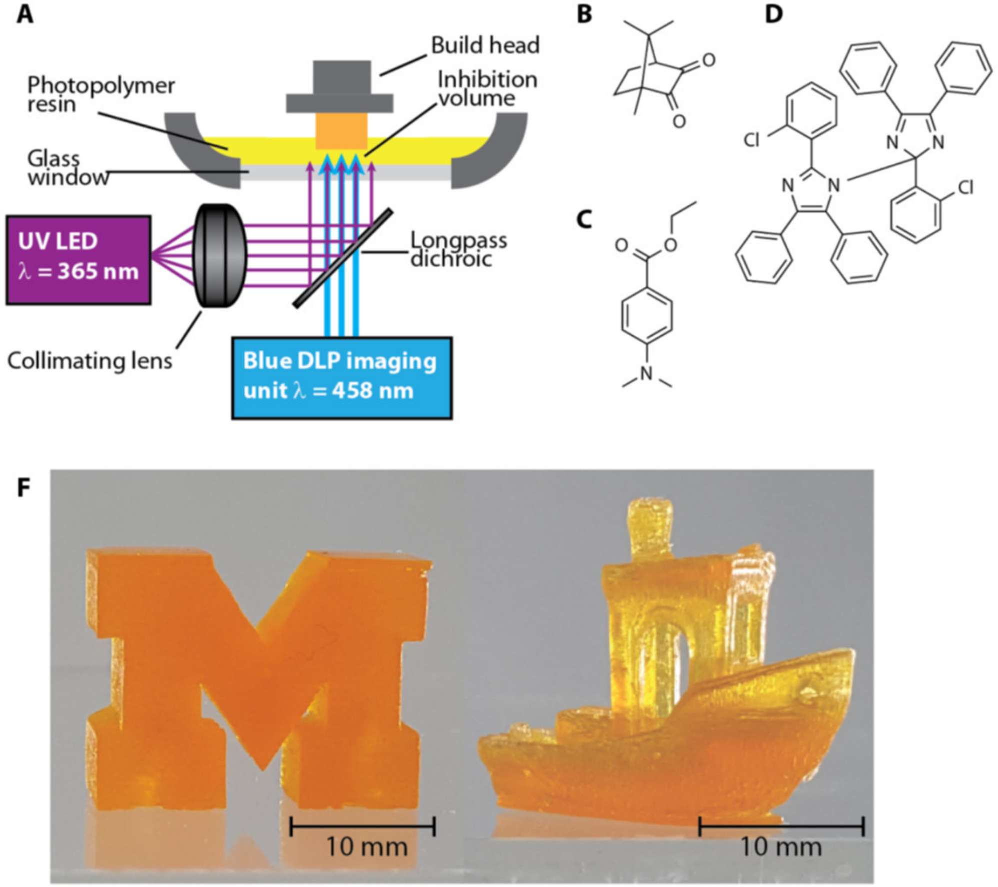
Top-Down



Lösung mit Monomer und Photoinitiator

Processes **2020**, *8*, 1138; doi:10.3390/pr8091138

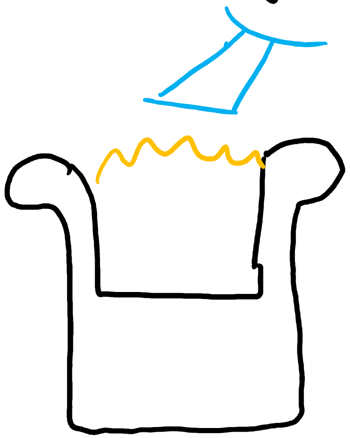
Photoinitiator +
Co-Initiator



Photoinhibitor

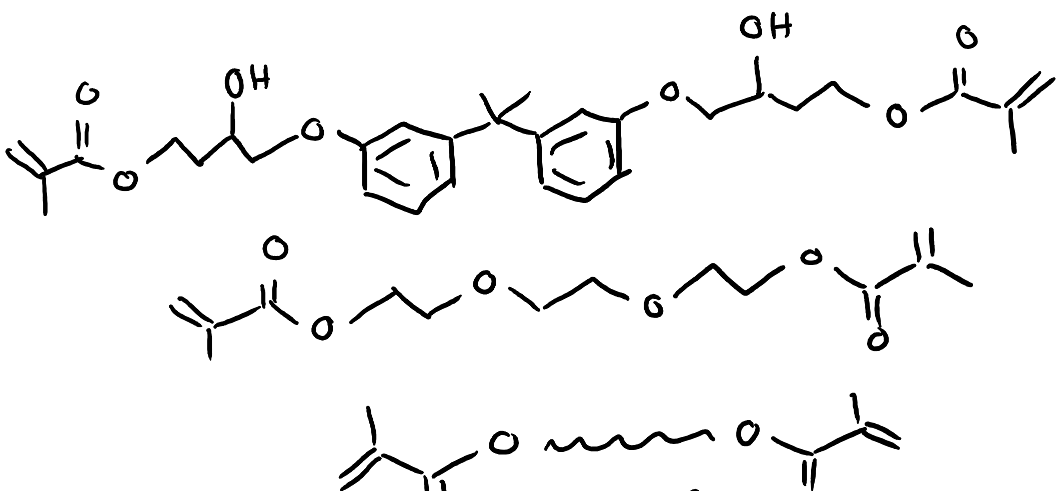
de Beer *et al.*, *Sci. Adv.* 2019; **5**:eaau8723

Beispiel Zahnfüllung



* mechanische Eigenschaften
* Quellung / Schrumpfung
* inert / Verträglichkeit
* Farbe
* Haftung

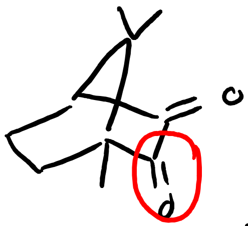
Beispiel für ein Rezept "Zahnfüllung"



Acryl

Mech. Stabilität

Schrumpfung! Quellung



Campherchinon

+ Co-Initiator

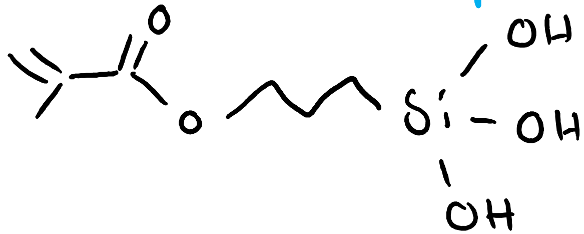
tert. Amine

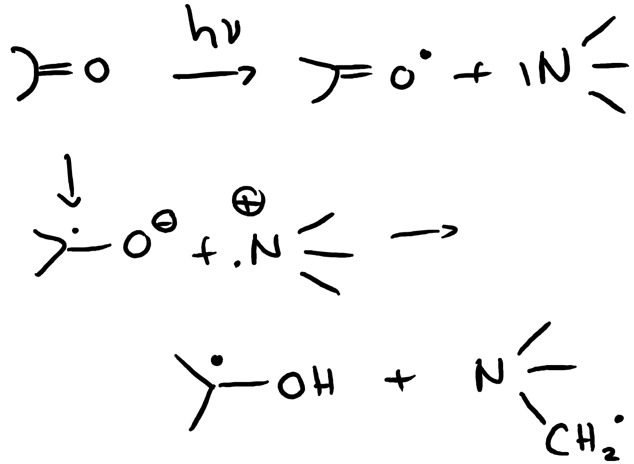
resin

filler : SiO_2

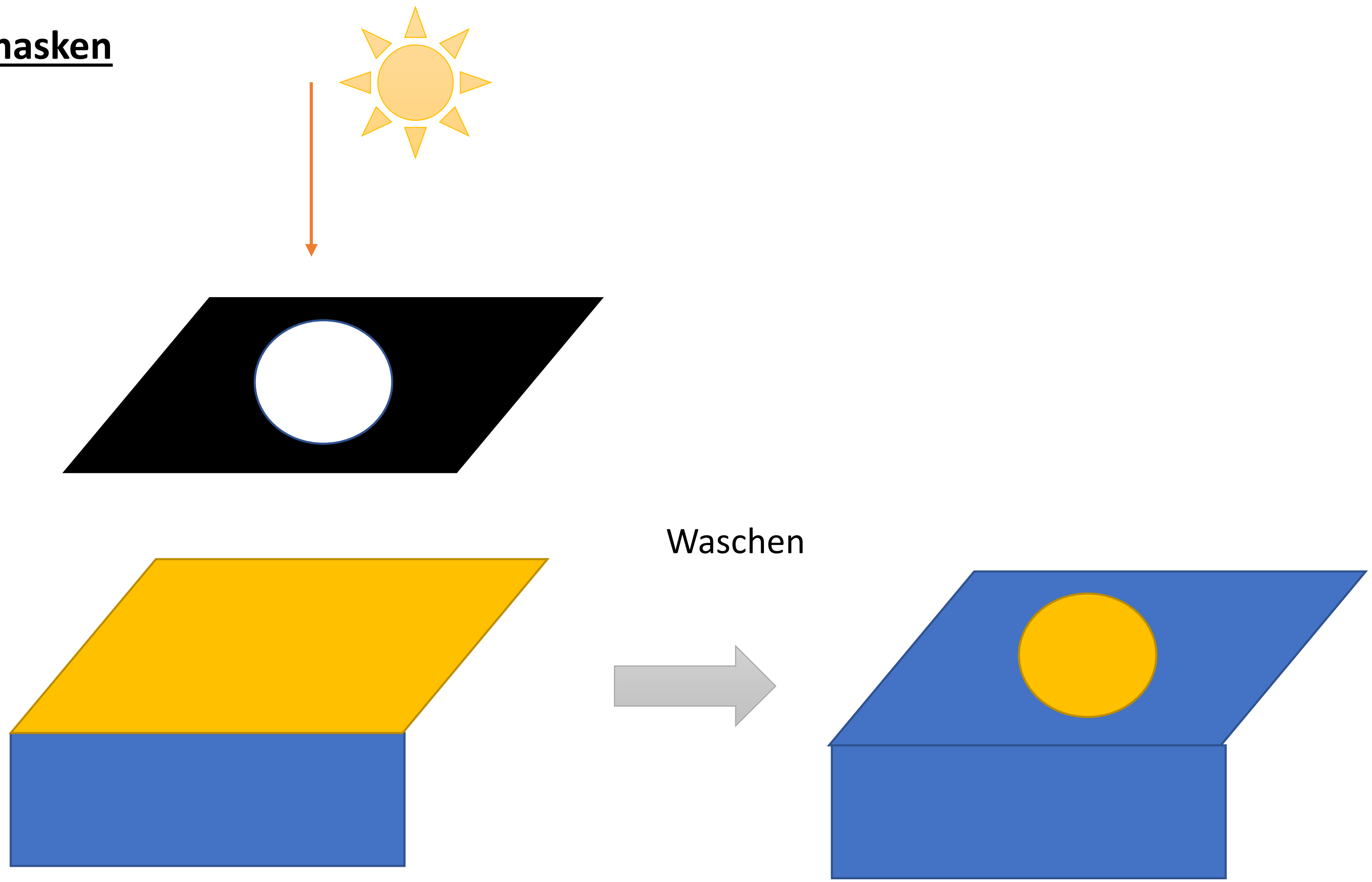
med. Stab.

Haftung

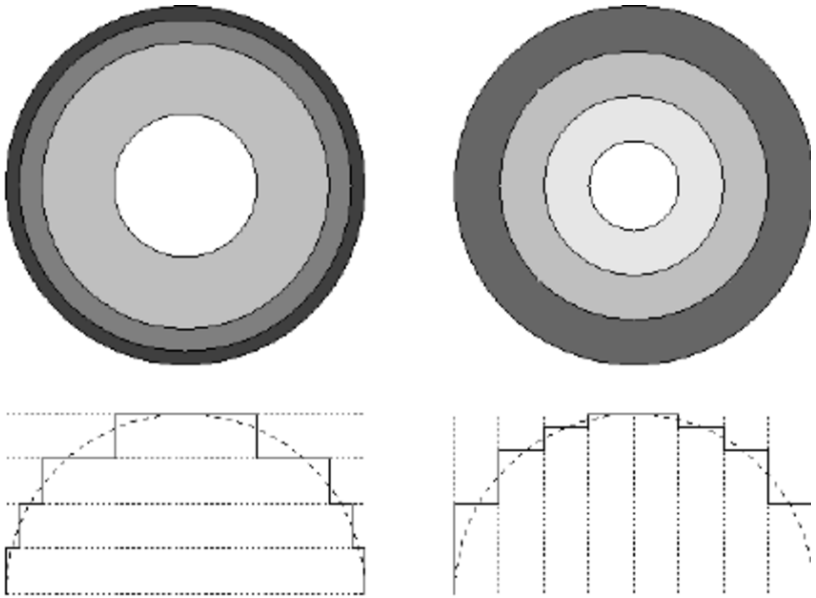




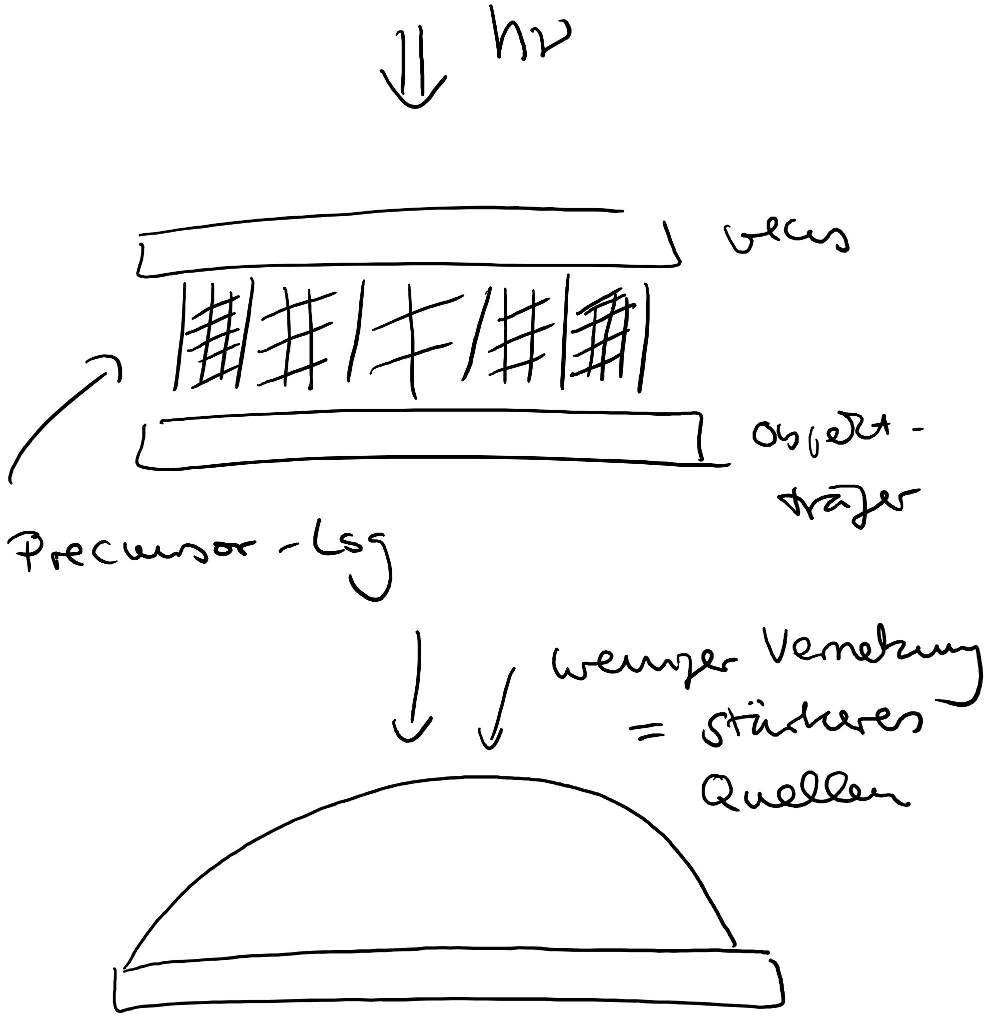
Photomasken



c) Photomaske



Uniform change in OD   Uniform zone size



Opt Express. 2004 Apr 5;12(7):1294-303.
Realization of refractive microoptics through grayscale lithographic patterning of photosensitive hybrid glass.
Rogers J, Kärkkäinen A, Tkaczyk T, Rantala J, Descour M.

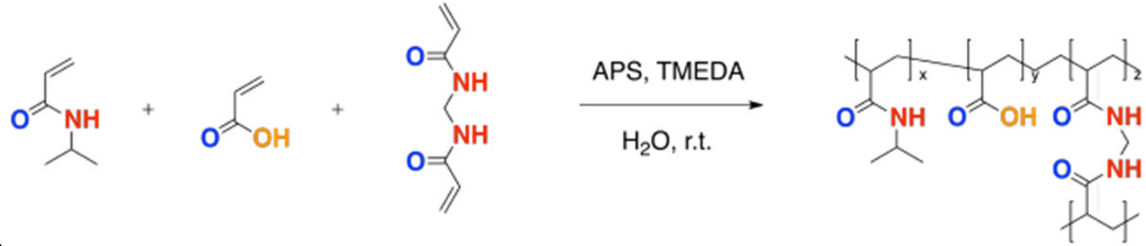
Anwendung von Hydrogelen

*Optical Control of Microfluidic Valves

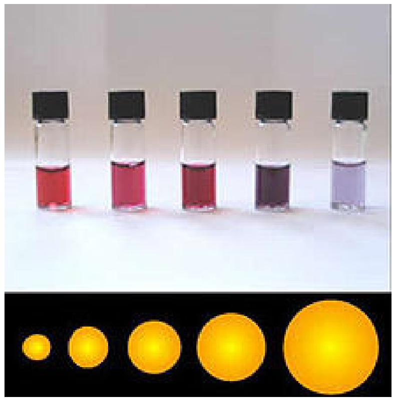
*Wundheilung

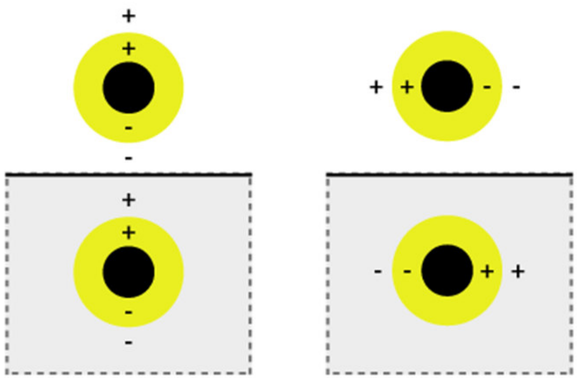
Optical Control of Microfluidic Valves

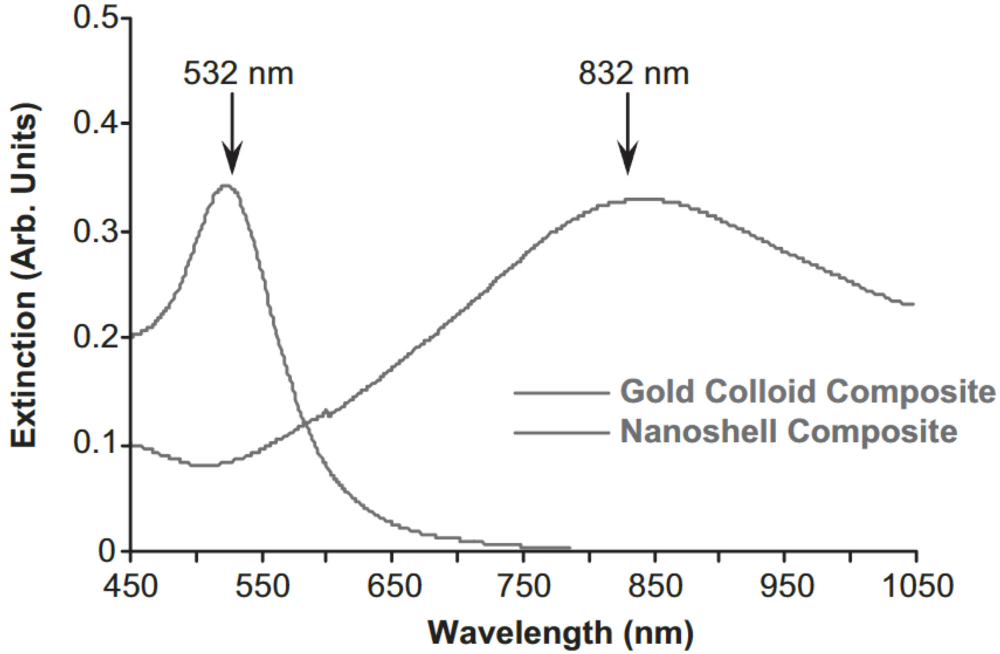
- (poly[N-isopropylacrylamide-co-acrylamide]



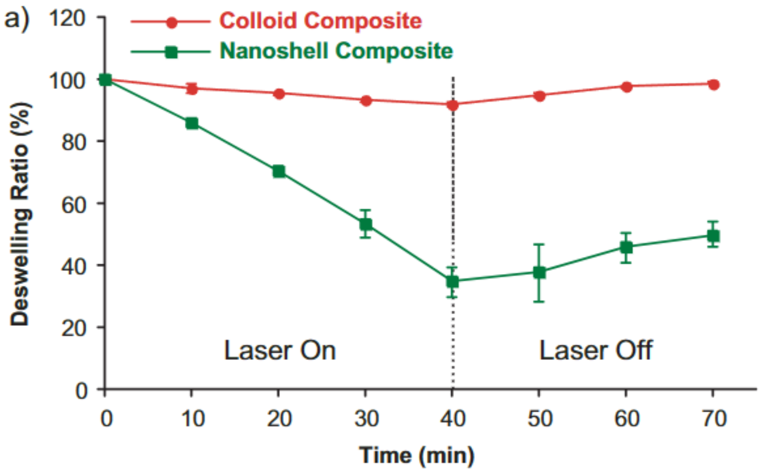
- Partikel mit unterschiedlichen starken optischen Absorptionsprofilen in Hydrogel eingeschlossen (gold colloids (2-100nm) and nanoshells (110 nm diameter silica core and 10 nm thick goldshell)





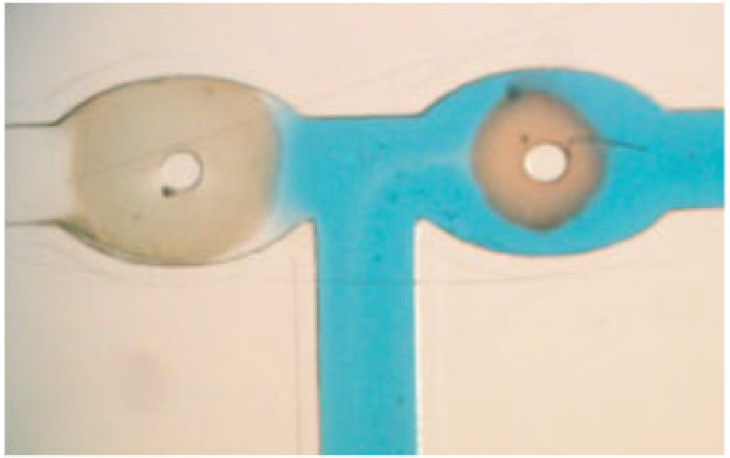


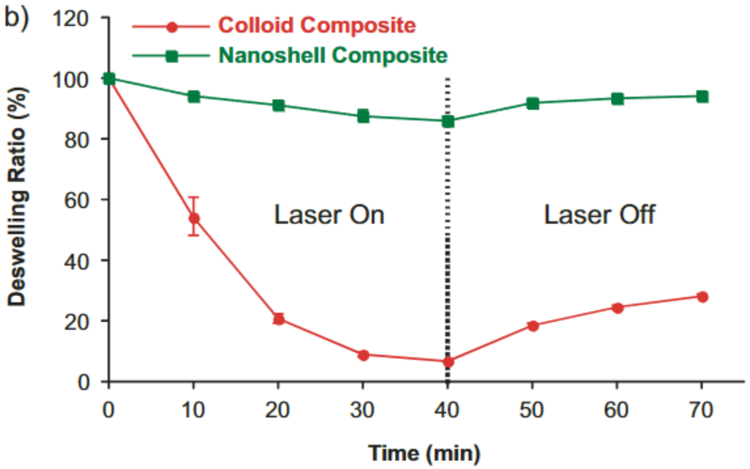
Optical Control of Microfluidic Valves



832 nm

a)

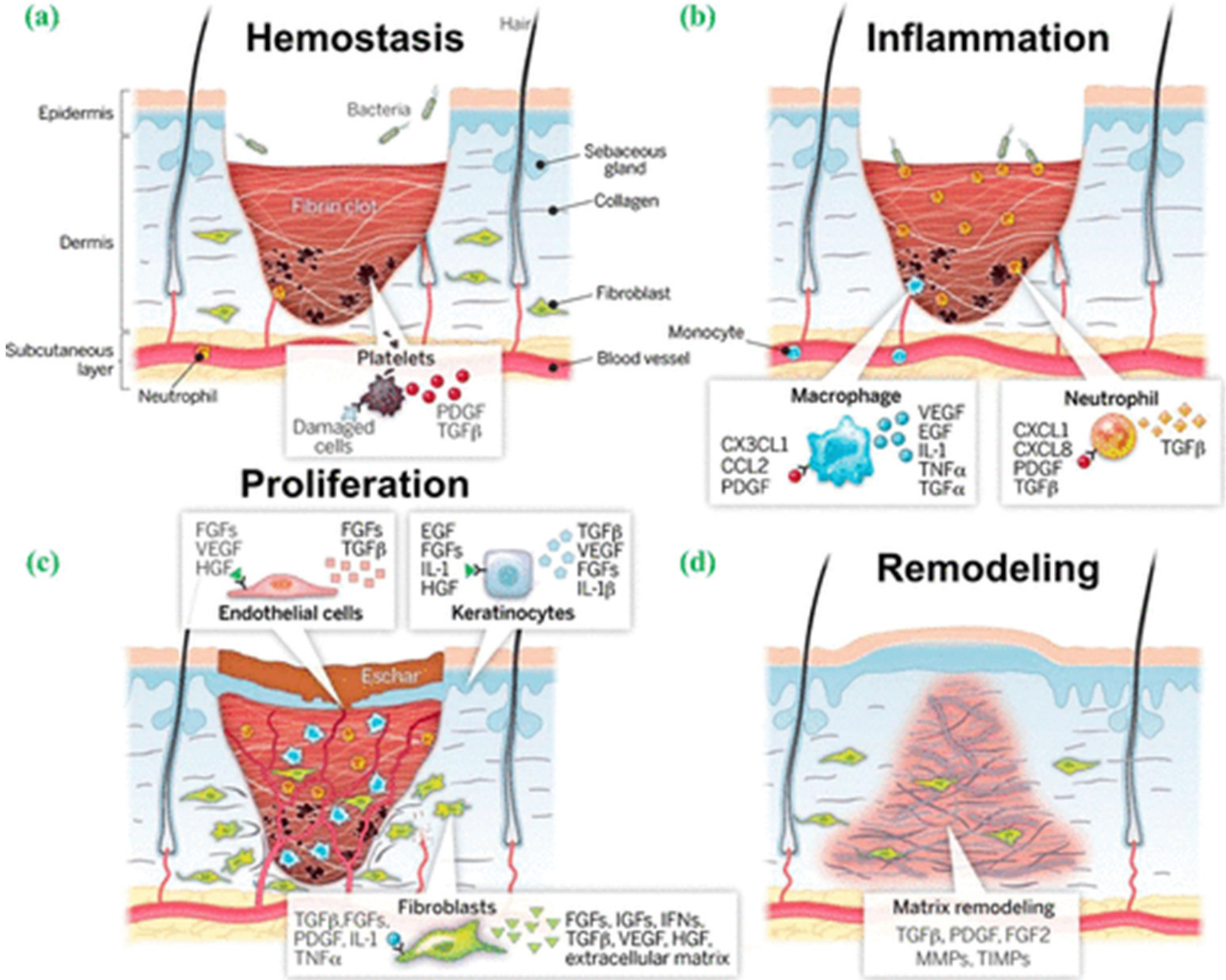




532 nm

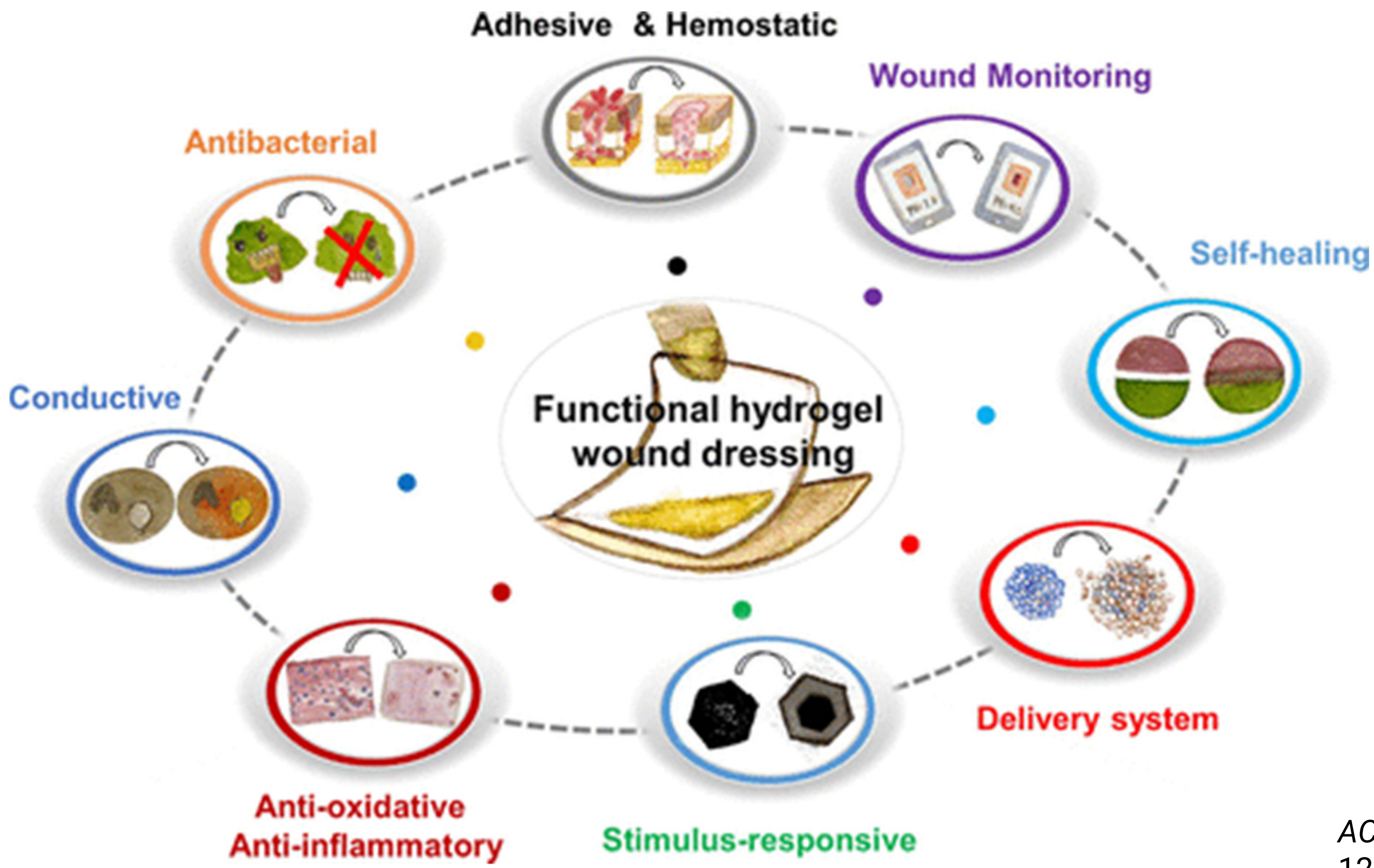
b)

Wundheilung



ACS Nano 2021, 15, 8, 12687–12722

Wundheilung



ACS Nano 2021, 15, 8, 12687–12722

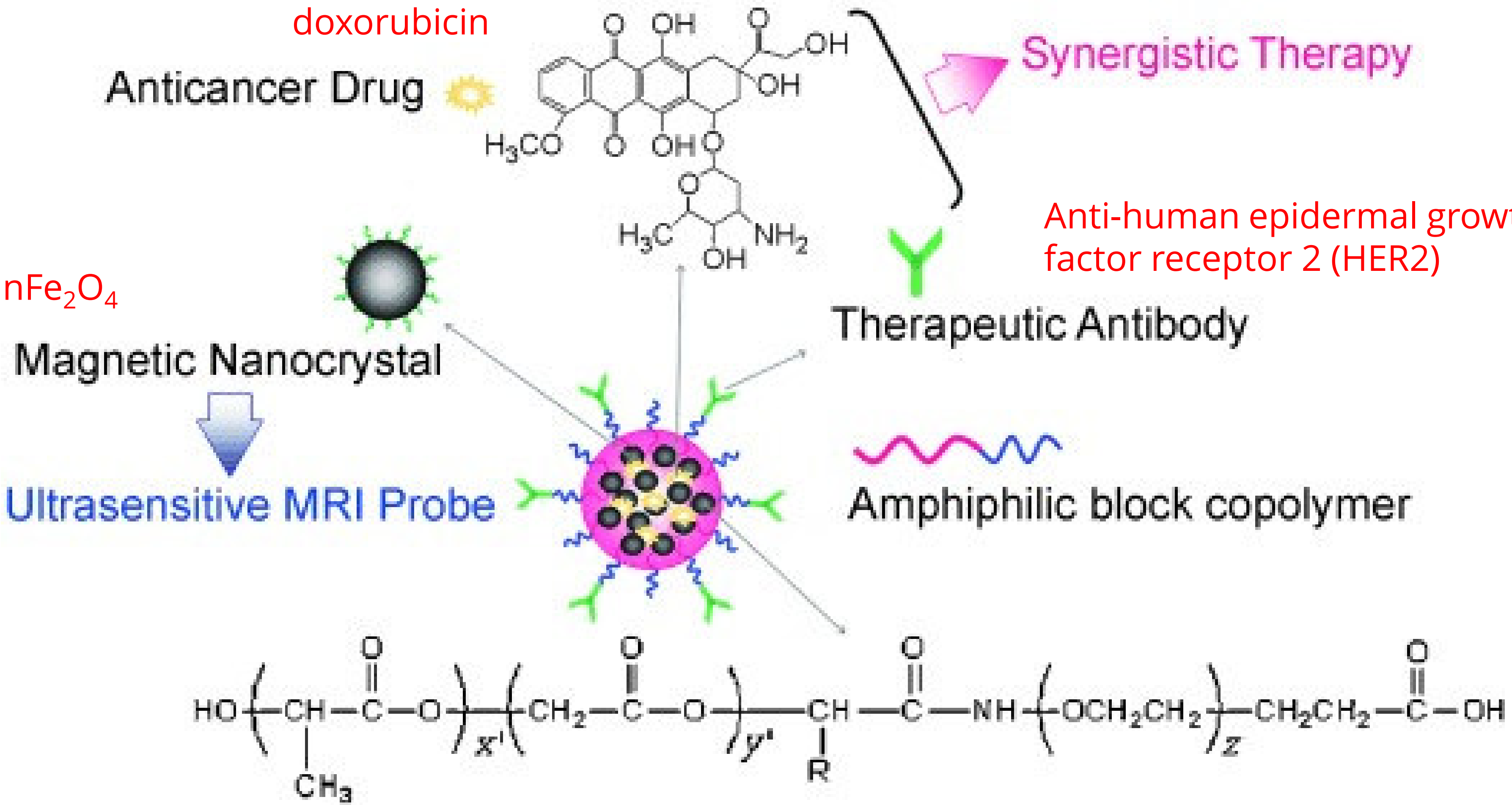
Anwendung von amphiphilen Block-Copolymeren

*MRI Probe

*Polymerosome als universelle carrier

Multifunctional Magneto-Polymeric Nanohybrids (MMPN)

doxorubicin

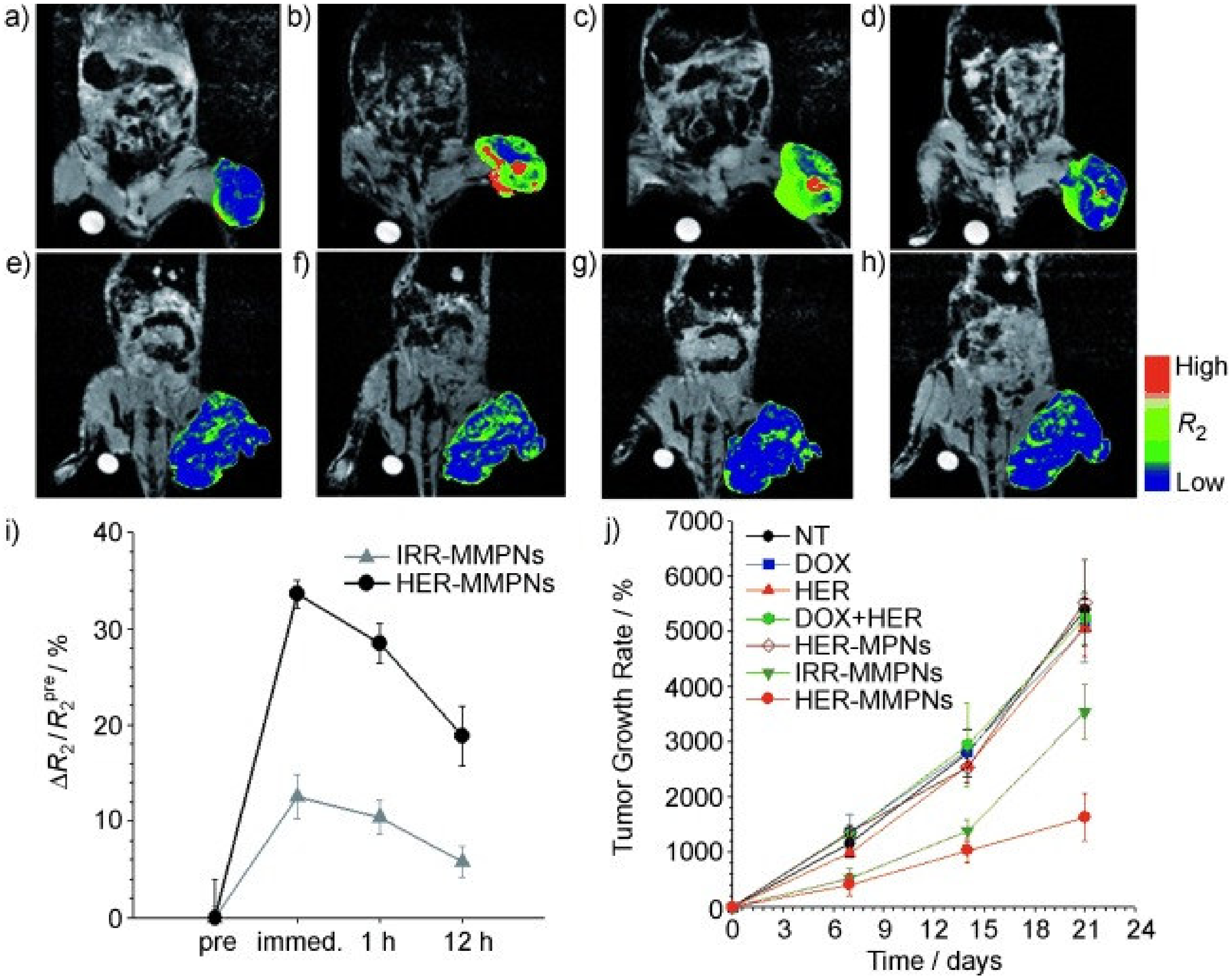


Anti-human epidermal growth factor receptor 2 (HER2)

$MnFe_2O_4$

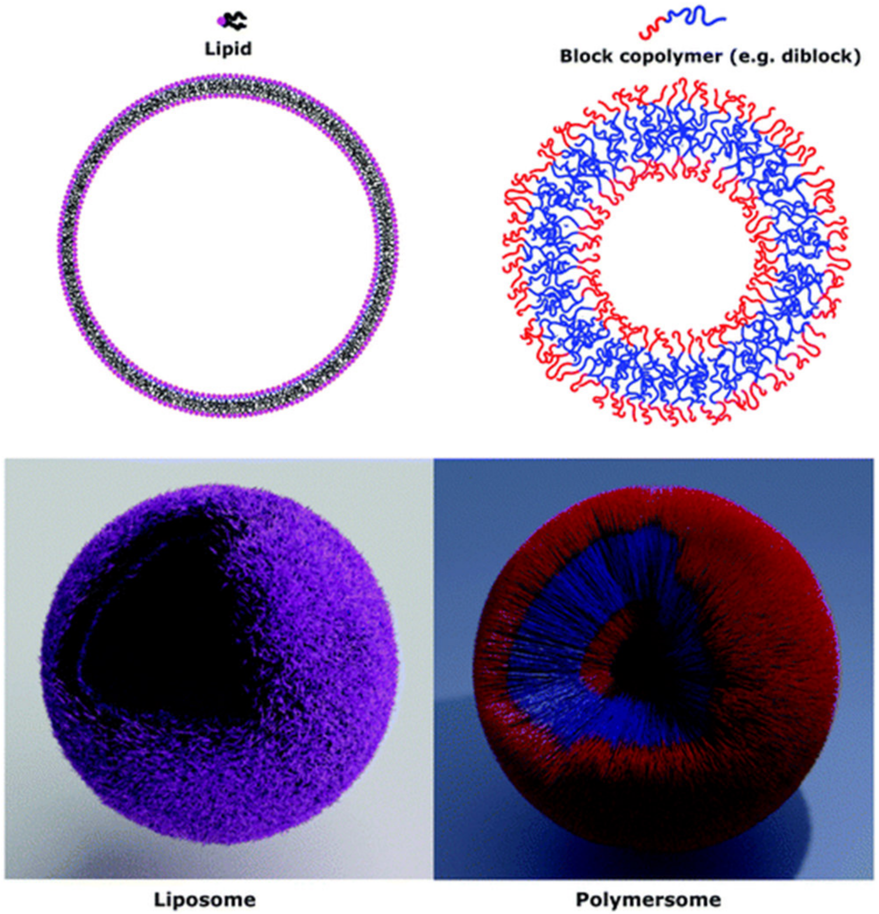
Angew Chem Int Ed **2007**, *46*, 8836-8839

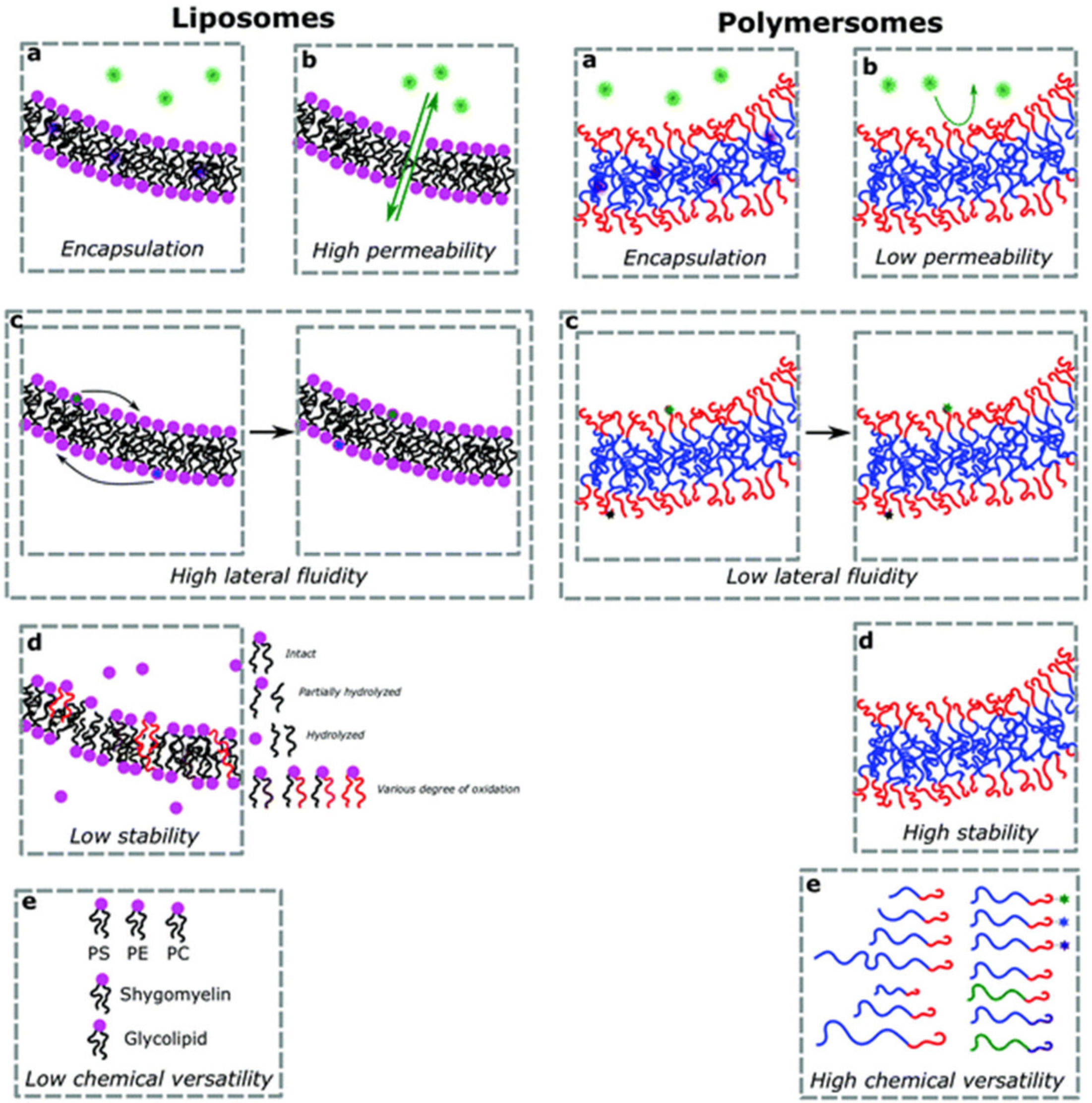
COOH-functionalized **Polylactide-*block*-poly(ethylene oxide)** = PLA-*b*-PEO irrelevant antibody (IRR, human IgG)



Angew Chem Int Ed **2007**, *46*, 8836-8839

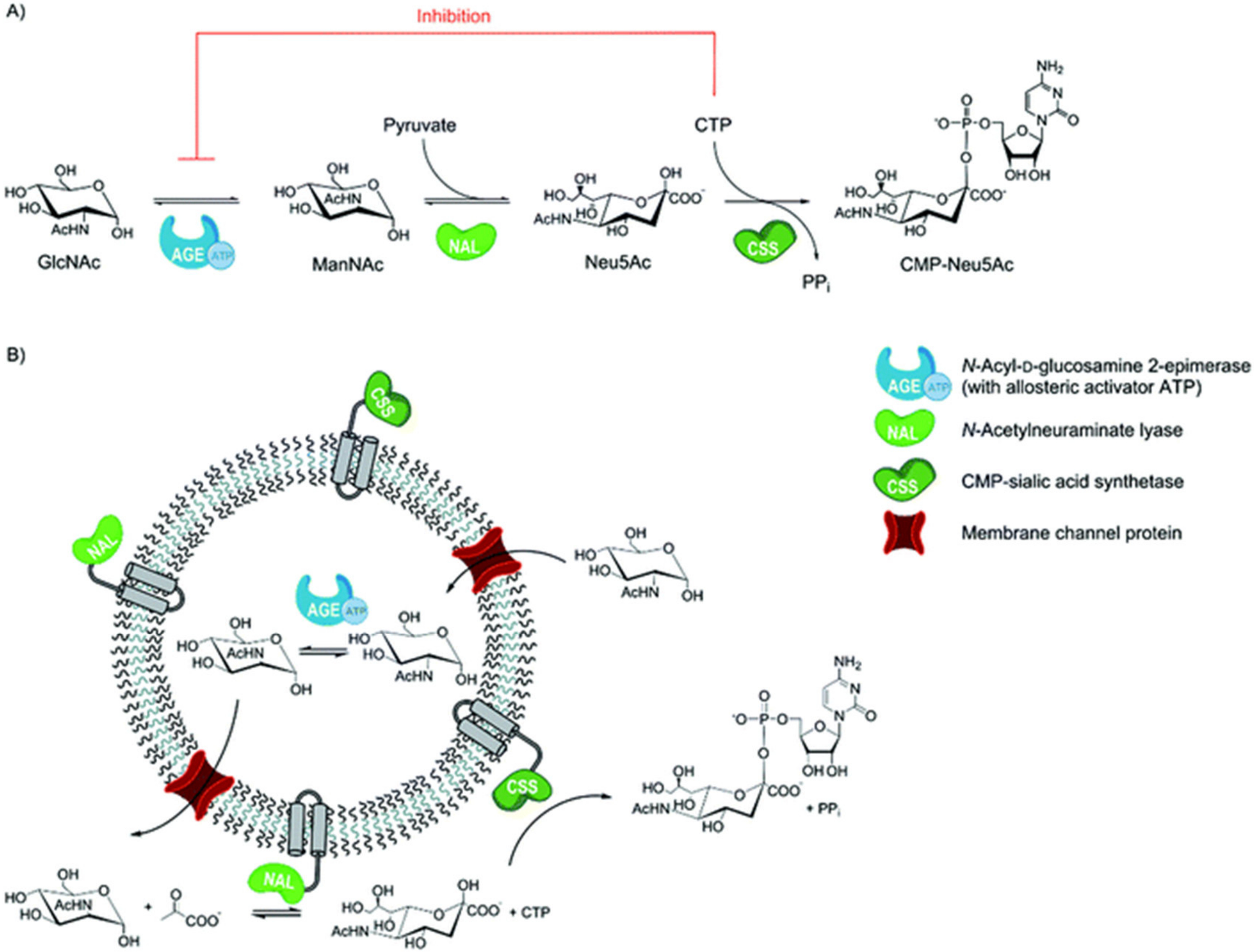
Polymerosome





Chem. Soc. Rev., 2018,**47**, 8572-8610.

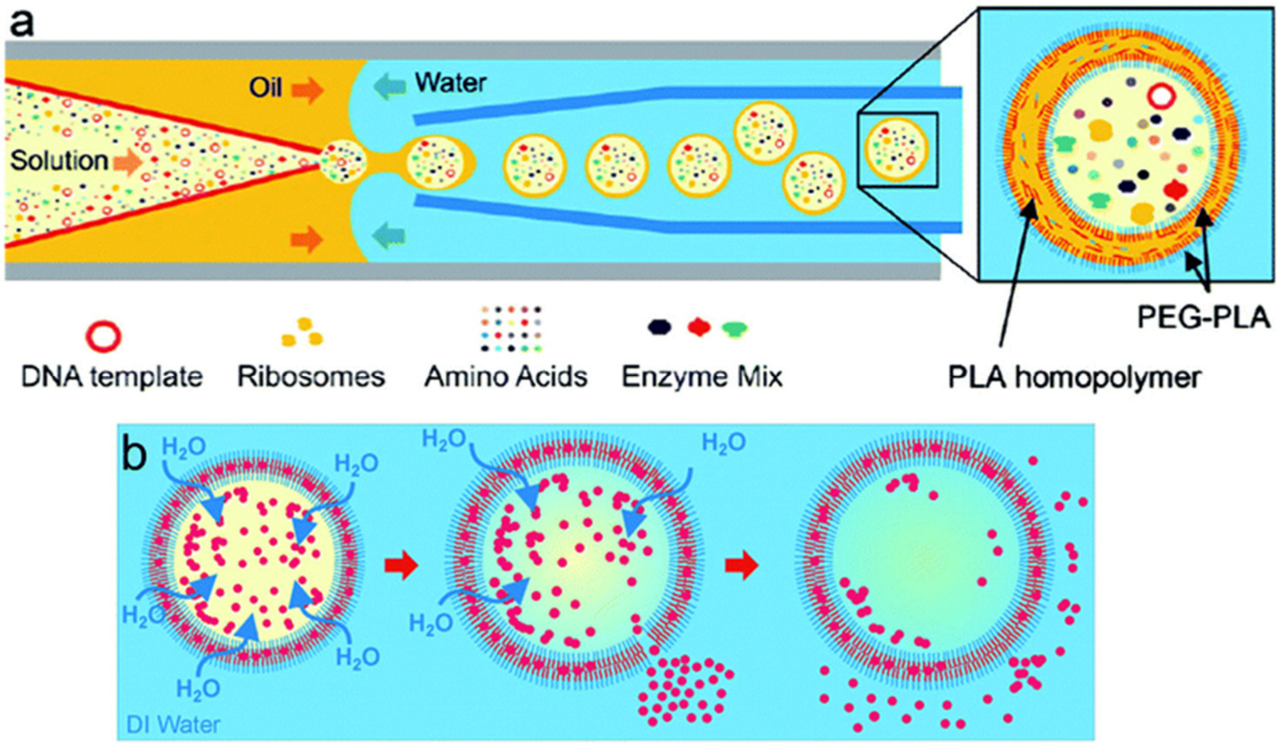
Polymerosome



- Multi-enzyme cascade carried out in a polymersome nanoreactor

ACS Catal., 2017, **7**(6), 3900–3904

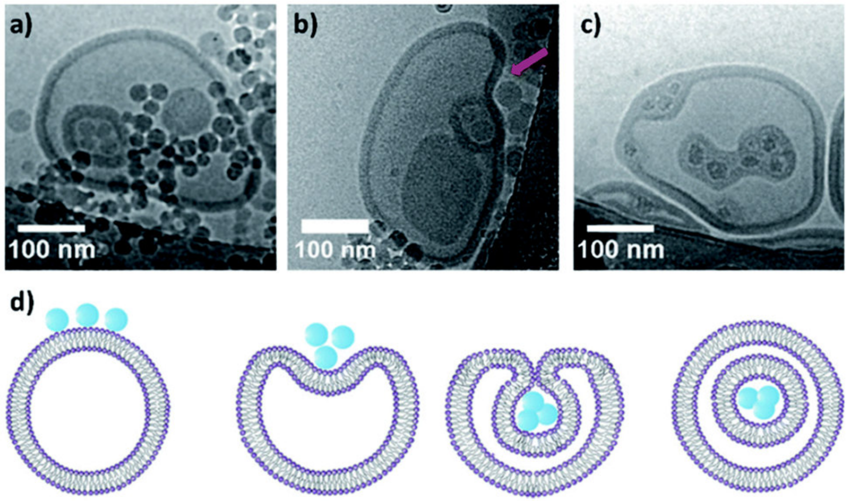
Polymerosome



➢ polymersome encapsulated an *E. coli* ribosomal expression (Cell free protein expression)

Angew. Chem., Int. Ed., 2012, **51**, 6416–6420

➢ encapsulation of PS and SiO_2 nanoparticles into polymersomes



Angew. Chem., Int. Ed., 2012, **51**, 4613–4617

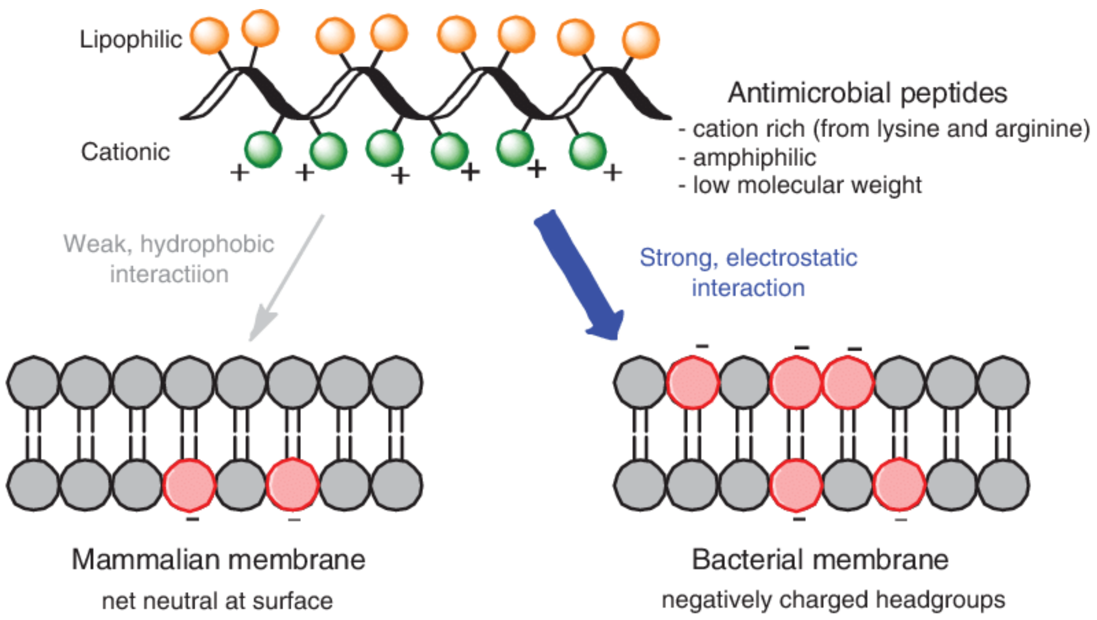
Anwendung von Miktoarmpolymeren

*Anti-Mikrobielle Reagenzien

*Dünnschichttechnologie

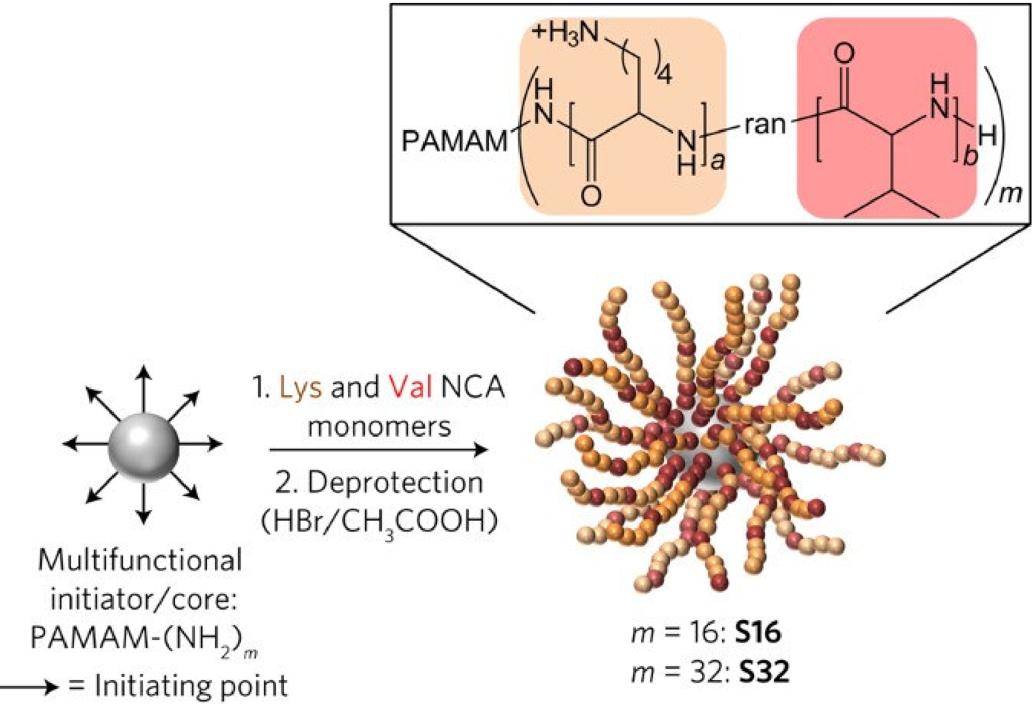
Anti-Mikrobielle Reagenzien

- Multi-resistente Bakterienstämme (Superbugs) sind ein wachsendes Problem im Gesundheitswesen (insbesondere in Krankenhäusern)



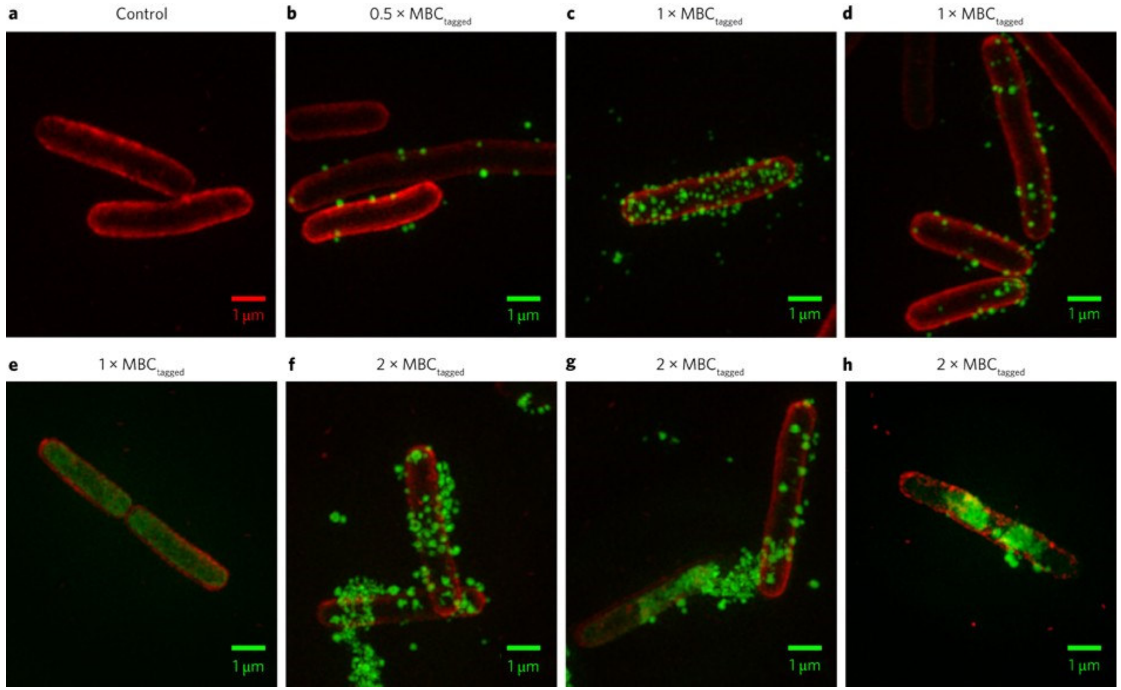
- Wirkungsweise AMPs

“structurally nanoengineered antimicrobial peptide polymers” (SNAPPs)



- ring-opening polymerization (ROP) of α-amino acid *N*-carboxyanhydrides (NCAs)

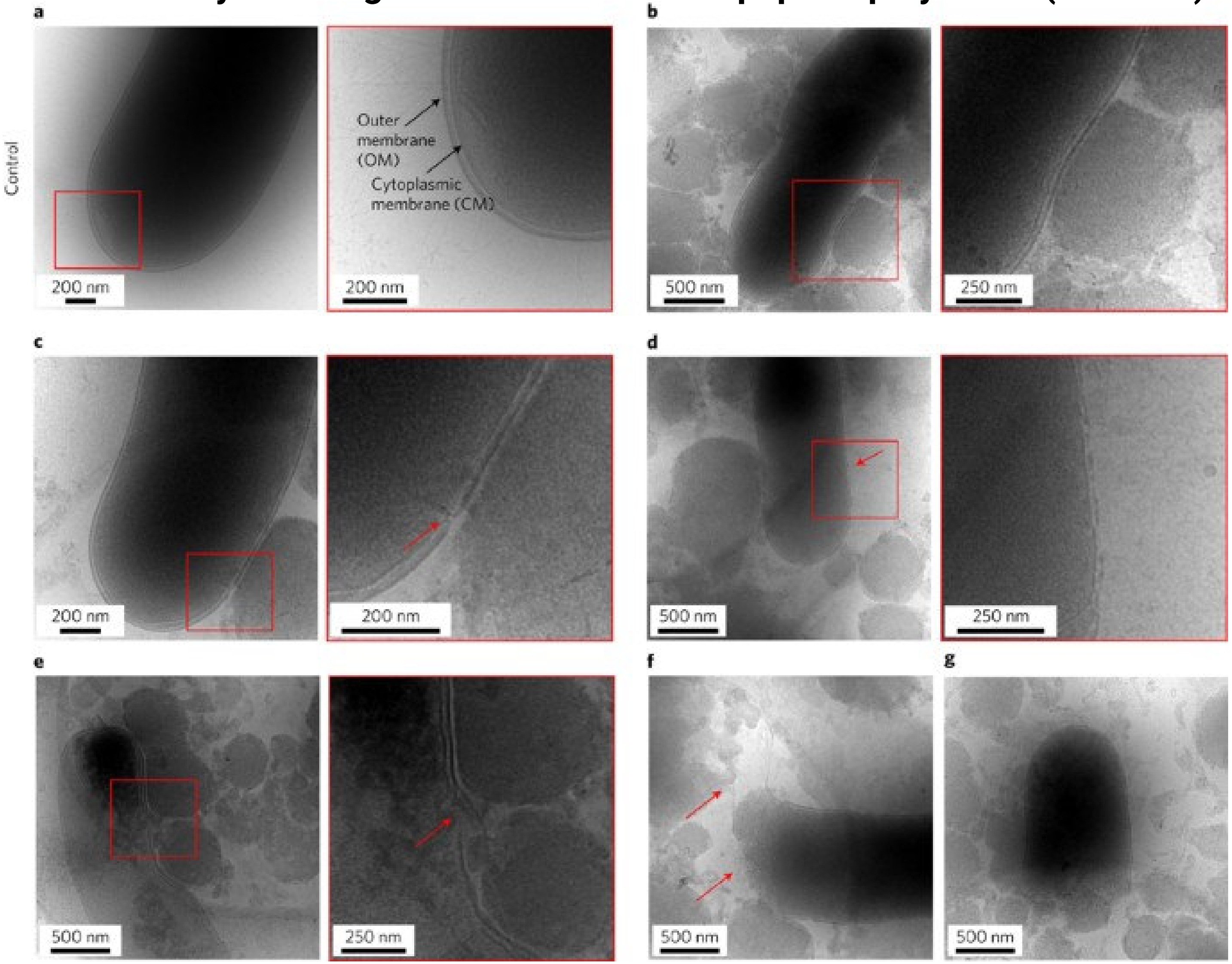
  - Anders als selbst-assemblierende antimikrobielle Makromoleküle, die unterhalb CMC in Monomere dissoziieren, bleiben SNAPPs stabile Architekturen auch in unendlich hoher Verdünnung



- *E. coli* (labelled red with lipid membrane FM4-64FX dye) incubated with the AF488-labelled **SNAPPs**

Nat. Microbio **2016**, *1* (11), 16162,

“structurally nanoengineered antimicrobial peptide polymers” (SNAPPs)

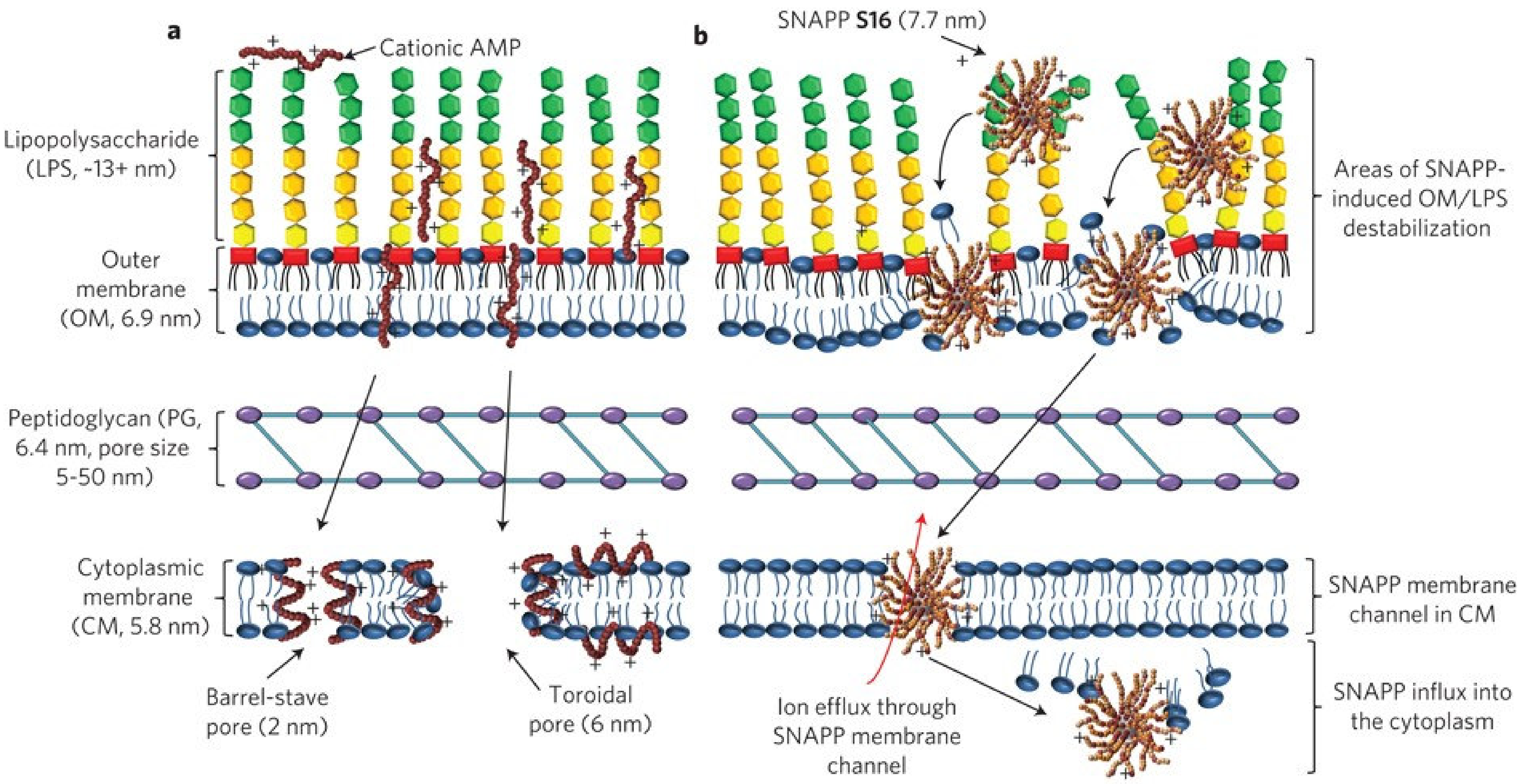


90 min at a lethal dose of 35 µg ml^{-1}

Multimodaler Mechanismus:

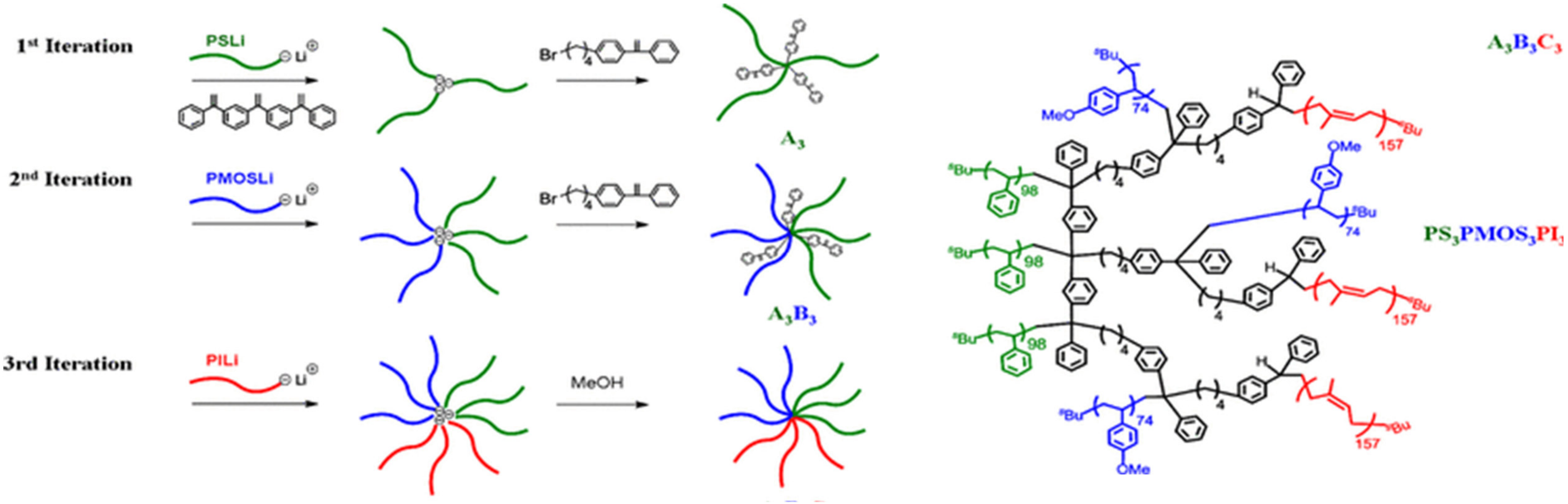
- Zustörung der OM
- Zerstörung der CM
- Ion efflux / influx
- Apoptose-ähnlicher Tod

“structurally nanoengineered antimicrobial peptide polymers” (SNAPPs)



Dünnschichttechnologie

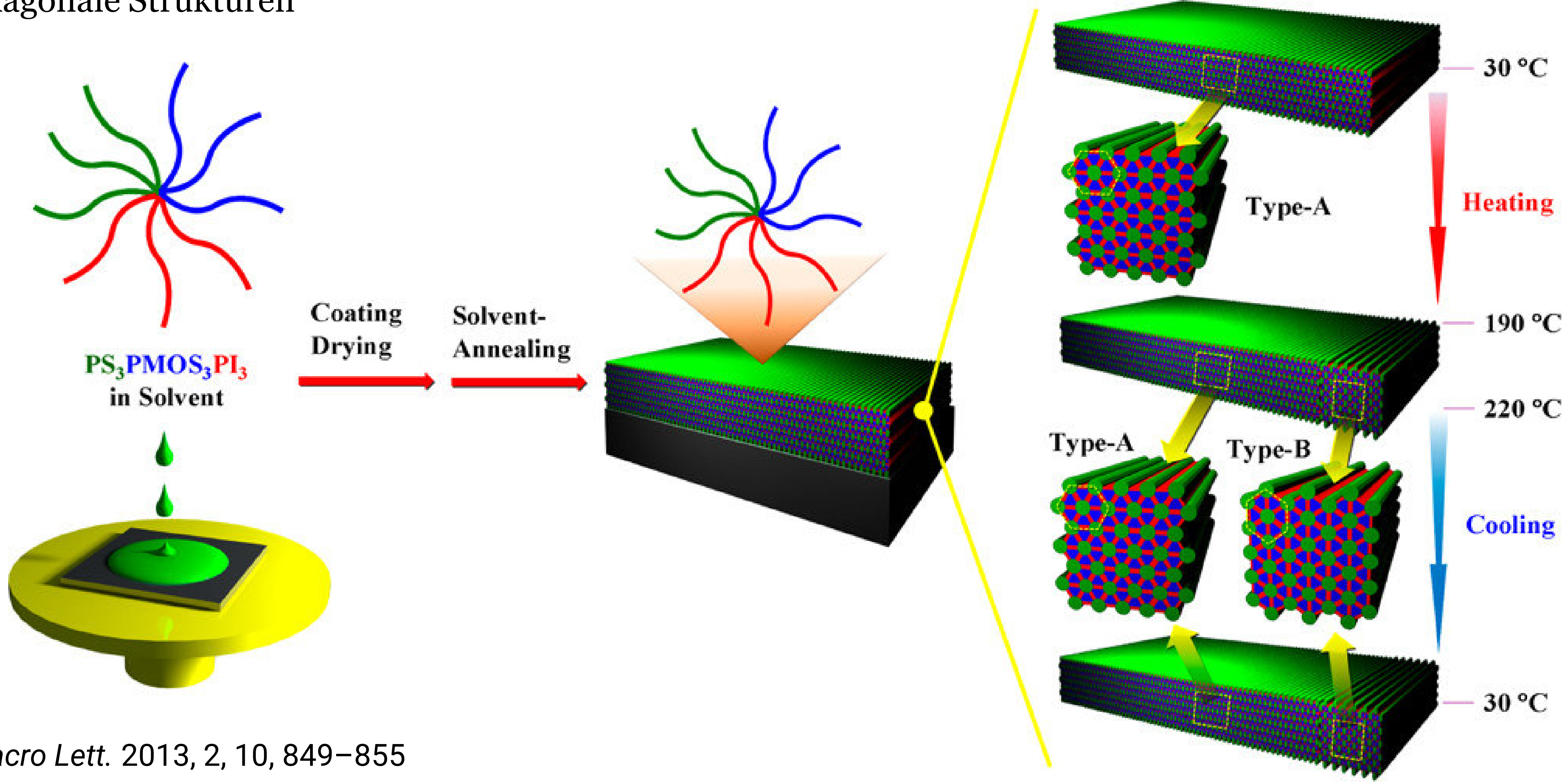
Synthese Strategie von (polystyrene)$_3$-(poly(4-methoxystyrene))$_3$-(polyisoprene)$_3$



ACS Macro Lett. 2013, 2, 10, 849–855

Dünnschichttechnologie

- Komplexe, hoch geordnete (*in-plane*)-ausgerichtete hexagonale Strukturen



ACS Macro Lett. 2013, 2, 10, 849–855